

FORT MACLEOD AGCM, AB, Canada

WMO: 717890

Lat: 49.7858N

Lon: 113.3794W

Elev: 978

StdP: 90.11

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-29.0	-25.4	-32.1	0.2	-28.8	-28.4	0.3	-25.4	18.4	3.0	16.5	3.6	2.4	260	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.2	30.6	15.7	28.6	15.3	26.8	14.9	18.1	25.3	17.1	25.1	16.1	24.1	5.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.5	12.4	21.6	14.2	11.4	20.3	13.0	10.5	18.5	55.4	25.1	51.7	25.1	48.6	24.1	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.0	13.3	11.6	DB	-35.0	34.3	3.0	2.5	-37.2	36.0	-39.0	37.5	-40.7	38.9	-42.9	40.6	(4)
(5)			WB	-35.1	20.7	3.0	1.2	-37.3	21.6	-39.0	22.3	-40.7	23.0	-42.9	23.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.3	-5.0	-6.3	-0.6	4.7	10.2	14.3	17.5	17.1	12.6	5.8	-0.5	-6.8	(6)	
	DBStd	10.71	8.48	8.95	7.65	5.02	4.26	3.15	2.71	3.24	4.62	5.69	7.72	9.05	(7)	
	HDD10.0	2487	466	455	330	168	49	3	0	1	26	149	318	521	(8)	
	HDD18.3	4804	724	688	587	410	252	127	47	61	176	388	564	779	(9)	
	CDD10.0	782	1	0	1	8	57	132	234	221	104	20	3	0	(10)	
	CDD18.3	57	0	0	0	0	1	5	23	23	5	0	0	0	(11)	
	CDH23.3	1439	0	0	0	8	74	134	506	524	189	4	0	0	(12)	
	CDH26.7	418	0	0	0	1	11	29	150	178	51	0	0	0	(13)	
(14) Wind	WSAvg	4.8	5.5	4.8	5.0	5.1	4.4	4.6	4.1	3.9	4.4	5.3	5.6	5.5	(14)	
(15) Precipitation	PrecAvg	424	18	15	26	37	56	76	44	46	42	18	18	21	(15)	
	PrecMax	612	48	34	77	121	184	215	137	121	146	53	54	82	(16)	
	PrecMin	276	2	1	4	3	24	7	4	0	4	1	0	2	(17)	
	PrecStd	85	13	9	16	30	34	50	35	32	32	16	15	17	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	13.2	16.4	24.3	28.2	30.8	33.0	33.6	30.7	23.1	18.1	10.9	(19)	
		MCWB	4.6	5.6	7.1	10.7	12.9	16.9	16.0	15.7	14.5	11.5	8.6	5.1	(20)	
	2%	DB	9.4	10.1	13.8	19.8	25.5	26.6	30.2	30.7	28.1	19.9	14.3	8.6	(21)	
		MCWB	4.1	4.0	6.1	8.7	12.6	14.8	16.1	15.6	14.0	9.9	7.5	3.5	(22)	
	5%	DB	7.4	7.7	11.8	16.7	22.6	24.5	28.1	28.5	25.5	17.1	11.6	6.5	(23)	
		MCWB	2.8	2.4	5.1	7.3	11.6	14.3	16.0	15.2	13.2	9.1	5.9	2.3	(24)	
	10%	DB	5.8	5.6	9.5	14.0	20.0	22.5	26.3	26.3	22.4	14.5	9.2	4.9	(25)	
		MCWB	1.9	1.4	3.8	6.0	10.4	13.3	15.9	14.7	12.3	7.5	4.4	1.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.8	6.3	8.0	11.1	14.2	18.5	20.6	18.3	16.0	12.8	9.5	5.8	(27)	
		MCDB	11.4	12.0	14.5	23.0	24.7	25.9	27.1	26.1	25.9	21.0	16.7	10.2	(28)	
	2%	WB	4.2	4.4	6.7	9.3	13.0	16.7	18.8	17.2	14.8	11.1	7.9	4.0	(29)	
		MCDB	8.9	9.7	13.0	18.2	23.9	24.7	25.6	26.5	25.4	18.7	14.2	7.6	(30)	
	5%	WB	3.2	2.8	5.2	7.7	12.0	15.2	17.7	16.4	13.7	9.4	6.1	2.7	(31)	
		MCDB	7.0	7.2	11.1	15.5	21.8	22.4	24.6	25.5	23.5	15.9	11.1	6.2	(32)	
	10%	WB	2.1	1.4	3.9	6.3	10.9	14.0	16.6	15.5	12.6	7.9	4.5	1.3	(33)	
		MCDB	5.7	5.4	8.9	13.2	18.9	20.9	23.9	24.3	21.6	14.1	9.0	4.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.6	11.6	11.6	13.0	13.4	13.2	15.2	15.8	14.6	12.5	10.9	10.5	(35)	
		MCDBR	10.5	13.2	14.5	18.6	19.1	17.7	19.7	20.8	20.9	15.9	12.3	10.6	(36)	
	5% WB	MCWBR	6.7	8.2	8.1	9.4	8.2	8.1	8.4	7.9	8.7	8.1	7.1	7.3	(37)	
		MCDWR	10.6	13.1	13.8	17.7	17.3	15.9	16.2	17.4	18.9	15.2	12.7	10.6	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.251	0.259	0.281	0.319	0.330	0.327	0.332	0.346	0.310	0.282	0.258	0.232	(40)	
		taud	2.435	2.467	2.479	2.388	2.417	2.471	2.468	2.410	2.484	2.537	2.499	2.448	(41)	
	Edn at Noon	Edn at Noon	826	910	941	928	928	930	920	889	893	868	813	806	(42)	
		Edh at Noon	59	74	88	108	110	105	104	105	87	69	56	50	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.24	2.23	3.50	4.84	5.77	6.40	7.03	5.59	3.97	2.47	1.37	0.97	(44)	
	RadStd	RadStd	0.10	0.18	0.29	0.33	0.47	0.46	0.36	0.35	0.34	0.26	0.11	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (67 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page